

A Scalar L^1 Counterexample to the BDP Converse for Lebesgue-Bochner Spaces

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Source Remark

The source paper is S. Basu and S. Seal, *Stability results of small diameter properties in Banach spaces*, arXiv:2011.14591. In the section “Lebesgue Bochner Function Spaces”, after Proposition `bochner bdp`, the authors write:

We do not know whether the converse of the Propositions `bochner bhp`, `bochner bdp` are true or not in general. However, converse of Proposition `bochner bhp`, fails for $p = 1$. Indeed, $\mathbb{R}$ has *BHP* but $L^1[0, 1]$ does not have *BHP*.

$$S(B_{L^p(\mu, X)}, f^*, \alpha) \cap S(B_{L^p(\mu, X)}, f^*, \alpha) = \emptyset.$$

$$a(S(B_{L^p(\mu, X)}, f^*, \alpha)) \geq \|g - h\|_p = \left(\sum_{i=1}^n \|y_i - z_i\|^p \mu(A_i) \right)^{1/p}$$

is a contradiction.

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 1. However, converse of Proposition 3.1, fails for $p =$
 μ is Lebesgue measure on $[0, 1]$ (which is simply denot

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quote the following result from [GGMS]

osition 4.1 [GGMS] X is strongly regular if and only if

Definitions

A Banach space X has the Ball Dentable Property (BDP) if, for every $\varepsilon > 0$, the unit ball B_X has a slice of diameter less than ε . It has the Ball Huskable Property (BHP) if B_X has a nonempty relatively weakly open subset of arbitrarily small diameter. Every slice is relatively weakly open, so

$$BDP \implies BHP.$$

Result

Theorem 1. *The converse of Proposition bochner bdp is false in general. More precisely, for Lebesgue measure on $[0, 1]$, the scalar space $X = \mathbb{R}$ has BDP, while $L^1([0, 1], X) = L^1[0, 1]$ does not have BDP.*

Proof. The one-dimensional space $\mathbb{R}$ has BDP: its unit ball has slices of arbitrarily small diameter.

It remains to show that $L^1[0, 1]$ does not have BDP. Let

$$S = \{f \in B_{L^1} : \Phi(f) > \|\Phi\| - \alpha\}$$

be a nonempty slice of $B_{L^1[0,1]}$, where $\Phi \in (L^1)^* = L^\infty$ and $\alpha > 0$. Choose $0 < \beta < \alpha$, and write $\varphi \in L^\infty[0,1]$ for the function representing Φ . The set

$$A = \{t \in [0,1] : |\varphi(t)| > \|\varphi\|_\infty - \beta\}$$

has positive measure. Since Lebesgue measure is non-atomic, A contains two disjoint measurable subsets A_1, A_2 of positive measure.

For $i = 1, 2$, define

$$f_i(t) = \operatorname{sgn}(\varphi(t)) \frac{\mathbf{1}_{A_i}(t)}{\mu(A_i)}$$

on A_i , with any measurable choice of sign on the null set where $\varphi = 0$. Then $\|f_i\|_1 = 1$ and

$$\Phi(f_i) = \frac{1}{\mu(A_i)} \int_{A_i} |\varphi(t)| d\mu(t) > \|\varphi\|_\infty - \beta > \|\Phi\| - \alpha.$$

Thus $f_1, f_2 \in S$. Their supports are disjoint, hence

$$\|f_1 - f_2\|_1 = \|f_1\|_1 + \|f_2\|_1 = 2.$$

Every slice of $B_{L^1[0,1]}$ therefore has diameter 2, so $L^1[0,1]$ does not have *BDP*. □

Consequence for the Source Remark

The source already observes that $X = \mathbb{R}$, $p = 1$, and Lebesgue measure on $[0,1]$ disprove the converse of Proposition `bochner bhp`. The theorem above shows that the same scalar example also disproves the converse of Proposition `bochner bdp`. Hence the two converse questions are both negative in the unrestricted “in general” sense.

Resolution History

No earlier partial packet for this exact result was found in the run. The active lane-17 triage first isolated the source remark, then checked the later paper [2]. That later paper proves Kothe-Bochner implications for *BDP* and *BHP*, and leaves a distinct *BSCSP* question open; it does not remove the $p=1$ scalar counterexample above. The packet directory contains a `history/` folder with the archived active-claim state and a short provenance note.

Scope and Verification

The argument is purely analytic and uses only the standard duality $(L^1)^* = L^\infty$ for finite measure spaces and the non-atomicity of Lebesgue measure. It does not address whether any positive converse holds for $1 < p < \infty$, for purely atomic measures, or under extra hypotheses on X .

The local run indexes were searched for arXiv:2011.14591, *BDP*, *BHP*, Lebesgue-Bochner, $L^p(\mu, X)$, and converse. The local arXiv corpus was also searched for the same phrases, including the later arXiv:2307.03631 Kothe-Bochner sequel. Web searches for exact phrases from the source remark and close variants returned the source paper and the 2023 sequel, but no separate explicit answer to this *BDP*-converse consequence.

References

- [1] S. Basu and S. Seal, *Stability results of small diameter properties in Banach spaces*, arXiv:2011.14591.
- [2] S. Basu and S. Seal, *Two aspects of small diameter properties*, arXiv:2307.03631.